



UNIVERSITY OF RWANDA

College of Science and Technology · Department of Applied Mathematics

A Fractional-Order Compartmental Model of HIV Transmission with Social Behaviour Interventions

NDACYAYISABA Lambert

Reg. 222008909 · B.Sc. Applied Mathematics

Supervisor: Dr. MUHIRWA Jean Pierre

Presentation content follows the submitted final-year project report

Final Year Project Report

Academic Year 2025–2026

Background / Motivation

Rwanda-aware public-health motivation

- ▶ HIV transmission is shaped by biology, treatment access, and social behaviour.
- ▶ Awareness, safer behaviour, testing, treatment-seeking, and adherence can change long-term outcomes.
- ▶ In Rwanda, continued prevention and adherence support remain important alongside treatment progress.

Why mathematics? A model gives a controlled way to compare intervention strategies and understand possible long-term behaviour.

Context

HIV in Rwanda

Mechanism

Social behaviour

Method

Fractional memory

Output

Simulation evidence

Problem Statement + Literature Gap

Limitations in ordinary modelling

- ▶ Ordinary models usually depend only on the present state.
- ▶ HIV-related behaviour may depend on past awareness, stigma, testing history, and adherence.
- ▶ Social behaviour is often simplified into fixed rates.

Literature gap

- ▶ Classical HIV models use ODE compartments such as SICA/SITA.
- ▶ Fractional models use Caputo derivatives for memory effects.
- ▶ Few works combine fractional HIV dynamics, behaviour controls, and an interactive dashboard.

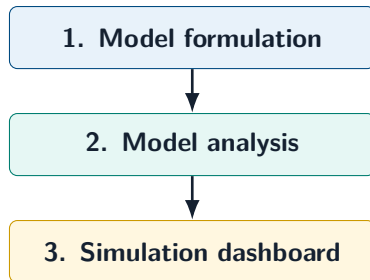
Research focus: a fractional-order SITA model that connects memory effects, social behaviour interventions, numerical simulation, and dashboard exploration.

Objectives

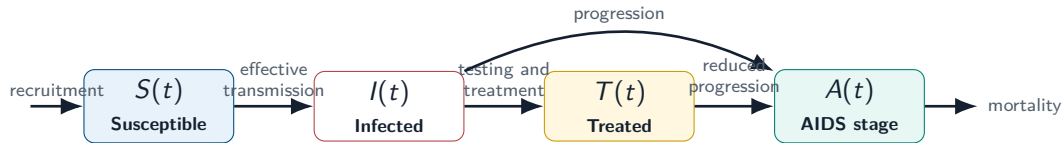
Main objective

To formulate, analyse, and computationally simulate a fractional-order SITA model of HIV transmission incorporating social behaviour interventions. **Specific objectives**

1. Formulate a fractional-order SITA HIV model using the Caputo derivative and social behaviour interventions.
2. Analyse positivity, boundedness, disease-free equilibrium, $\mathcal{R}_0$, and fractional stability.
3. Implement simulations and a Python dashboard to study memory effects, interventions, sensitivity, and epidemic trajectories.



Model Diagram: SITA



State variables: $S(t)$ susceptible, $I(t)$ infected, $T(t)$ treated, $A(t)$ AIDS-stage population.

Controls: u_1 awareness, u_2 safer behaviour, u_3 testing, u_4 adherence.

Fractional-Order SITA Model

Objective 1: model formulation

$${}^C D_t^q X(t) = F(t, X(t)), \quad 0 < q \leq 1$$

$${}^C D_t^q S = \Lambda - \lambda(t)S - \mu S,$$

$${}^C D_t^q I = \lambda(t)S - (\tau_{\text{eff}} + \delta + \mu)I,$$

$${}^C D_t^q T = \tau_{\text{eff}}I - (\rho_{\text{eff}} + \mu)T,$$

$${}^C D_t^q A = \delta I + \rho_{\text{eff}}T - (d + \mu)A.$$

$$\lambda(t) = \frac{\beta_{\text{eff}}(I + \eta T)}{N},$$

$$\beta_{\text{eff}} = \beta_0(1 - u_1)(1 - u_2), \quad \tau_{\text{eff}} = \tau(1 + u_3),$$

$$\rho_{\text{eff}} = \rho(1 - u_4).$$

Meaning of interventions

u_1, u_2 reduce effective transmission through awareness and safer behaviour.

u_3 increases testing/treatment movement from infected to treated.

u_4 reduces progression from treatment to AIDS stage through adherence support.

When $q = 1$ the model becomes ordinary; when $q < 1$ previous states influence the present trajectory.

Objective 2: Mathematical Analysis Evidence

Positivity proof

$$\begin{aligned} S = 0 : {}^C D_t^q S &= \Lambda \geq 0, & I = 0 : {}^C D_t^q I &= \lambda(t)S \geq 0, \\ T = 0 : {}^C D_t^q T &= \tau_{\text{eff}} I \geq 0, & A = 0 : {}^C D_t^q A &= \delta I + \rho_{\text{eff}} T \geq 0 \end{aligned}$$

Therefore the vector field points inward on the boundary, so the nonnegative region $\mathbb{R}_+^4$ is positively invariant.

Boundedness proof

$$N = S + I + T + A, \quad {}^C D_t^q N = \Lambda - \mu N - dA \leq \Lambda - \mu N.$$

$$\limsup_{t \rightarrow \infty} N(t) \leq \frac{\Lambda}{\mu}.$$

Disease-free equilibrium and threshold

$$E_0 = \left(\frac{\Lambda}{\mu}, 0, 0, 0 \right), \quad \mathcal{R}_0 = \rho(FV^{-1}).$$

F contains new-infection terms and V contains transition/removal terms of the infected subsystem. Thus $\mathcal{R}_0$ is obtained by the next-generation matrix method.

Fractional stability condition

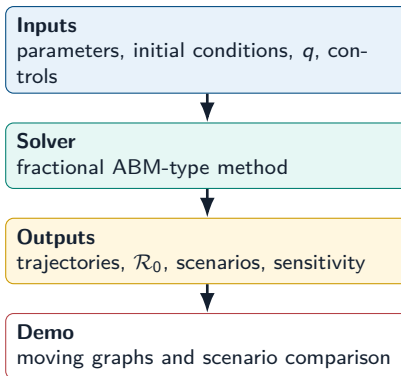
$$|\arg(\lambda_i)| > \frac{q\pi}{2} \quad \text{for every eigenvalue } \lambda_i.$$

When $\mathcal{R}_0 < 1$ and the eigenvalues satisfy this Matignon condition, the disease-free equilibrium is locally asymptotically stable.

Numerical Scheme + Implementation

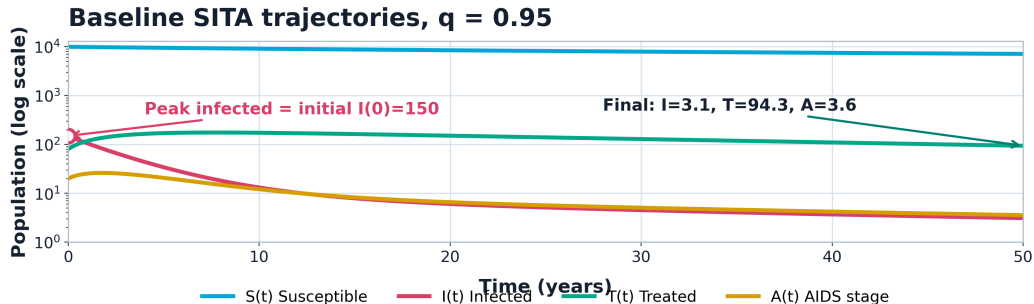
Numerical idea

- ▶ Time interval is divided into discrete steps.
- ▶ A predictor estimates the next state.
- ▶ A corrector updates the state using fractional-memory weights.
- ▶ One engine supports baseline, memory, intervention, and sensitivity experiments.



Presentation link: the slides explain the model; moving graphs are shown separately during the live demonstration.

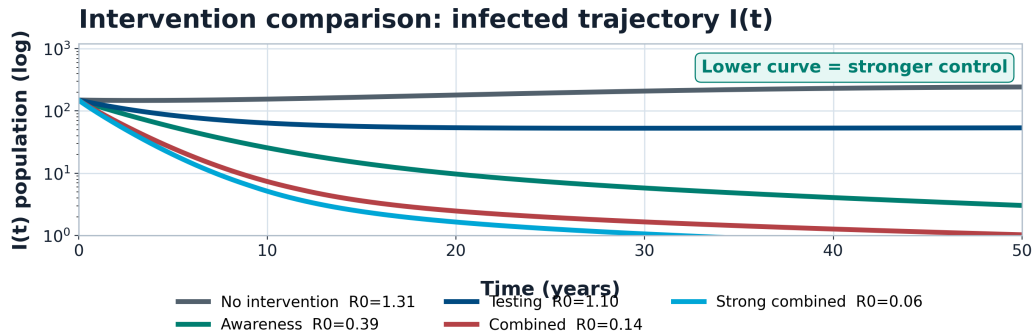
Baseline Simulation Result



► $\mathcal{R}_0 = 0.430 < 1$. The infected class starts at its maximum $I(0) = 150$ and then declines to $I = 3.1$, while treatment remains visible at $T = 94.3$. The log scale lets the audience see both the large susceptible class and the smaller infected/AIDS curves.

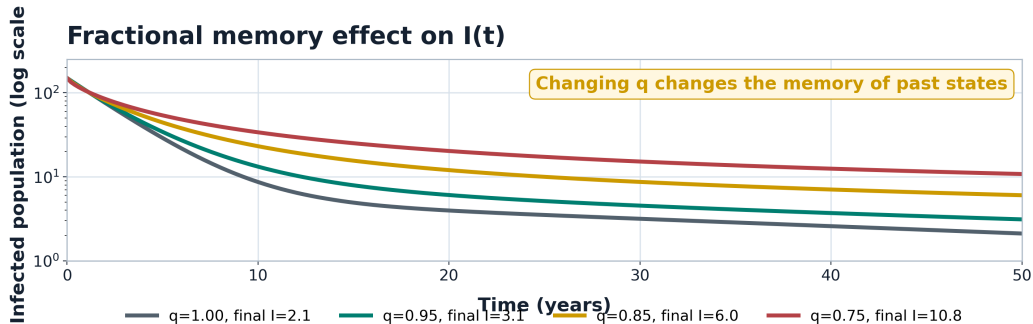
Source: submitted report, Chapter 6 baseline fractional simulation.

Simulation Results: Intervention Comparison



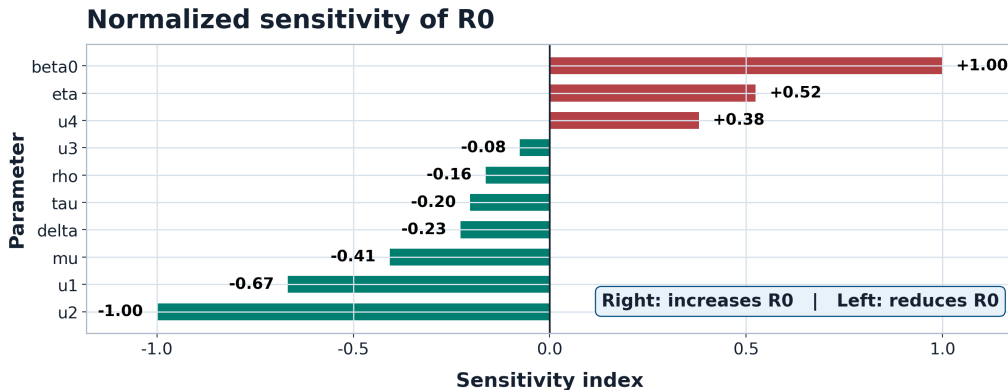
► Curves lower on the graph indicate fewer infected individuals. Single controls improve outcomes, but combined strategies reduce transmission, improve treatment movement, and slow AIDS progression together.
Source: submitted report, Chapter 6 intervention comparison.

Simulation Results: Fractional Memory



► $q = 1$ is the ordinary model. Values $q < 1$ introduce memory, so past states continue to influence the present trajectory. The separation between the curves shows the computational effect of fractional memory.
Source: submitted report, Chapter 6 fractional-memory comparison.

Dashboard and Sensitivity Evidence



► Normalized index: $\Upsilon_p^{\mathcal{R}_0} = \frac{\partial \mathcal{R}_0}{\partial p} \frac{p}{\mathcal{R}_0}$. Positive bars increase $\mathcal{R}_0$; negative bars reduce $\mathcal{R}_0$. Transmission raises epidemic potential, while behaviour controls push $\mathcal{R}_0$ downward.
Source: submitted report, Chapter 6 sensitivity analysis.

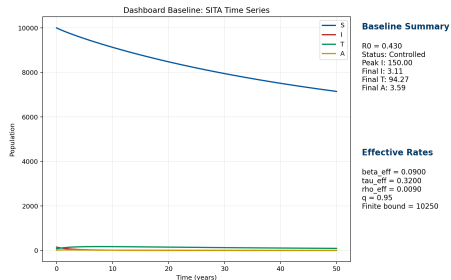
Live Dashboard Demonstration

Open during the defense

Dashboard link

<https://hiv-model.onrender.com/>

1. Run the baseline simulation.
2. Show the final state summary and interpretation.
3. Compare no intervention with combined intervention.
4. Change q to show the fractional memory effect.



Conclusion + Recommendations

Conclusion

- ▶ The SITA HIV model was extended using fractional memory.
- ▶ Behavioural interventions were represented through effective rates.
- ▶ Numerical simulations support the value of combined intervention strategies.

Recommendations

- ▶ Calibrate parameters with Rwanda-specific data.
- ▶ Extend the model using age, gender, or stochastic structure.
- ▶ Compare other fractional operators.
- ▶ Improve the dashboard into an educational decision-support tool.

Final message: the work connects mathematical analysis, numerical simulation, and a working dashboard for communicating HIV intervention dynamics.